Continuity

Section 1: Definition of Continuity at a Point

1 Which three conditions must be satisfied for a function f to be continuous at $x = c$?

a. $f(c)$ is defined, $\lim_{x \to c} f(x)$ exists, and $\lim_{x \to c} f(x) = f(c)$

b. $f'(c)$ exists, $f(c) > 0$, and $\lim_{x \to c} f(x) = 0$

c. $\lim_{x \to c^-} f(x) = \lim_{x \to c^+} f(x)$ only

d. $f(c) = 0$ and $f'(c) = 0$

2 Determine if $f(x)$, defined as $\frac{x^2 - 4}{x - 2}$ for $x \neq 2$ and $f(2) = 5$, is continuous at $x = 2$.

a. Discontinuous (removable discontinuity at $x = 2$)

b. Continuous at $x = 2$

c. Discontinuous (jump discontinuity)

d. Discontinuous (infinite discontinuity)

3 For what value of a is $h(x)$, defined as $ax^2 + 1$ for $x \leq 2$ and $3x - a$ for $x > 2$, continuous everywhere?

a. 1

b. 2

c. $3/5$

d. -1

Section 2: Classification of Discontinuities

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

4 Classify the discontinuity of $g(x) = \frac{x-1}{|x-1|}$ at $x = 1$.

a. Jump discontinuity

b. Removable discontinuity

c. Infinite discontinuity

d. Essential oscillating discontinuity

5 Classify the discontinuity of $f(x) = \frac{1}{(x-4)^2}$ at $x = 4$.

a. Infinite discontinuity

b. Removable discontinuity

c. Jump discontinuity

d. Continuous

6 How can $f(x) = \frac{\sin(3x)}{x}$ be defined at $x = 0$ to make it continuous?

a. Set $f(0) = 3$

b. Set $f(0) = 1$

c. Set $f(0) = 0$

d. It cannot be made continuous

Section 3: Intermediate Value Theorem (IVT)

7 State the primary hypothesis required to apply the Intermediate Value Theorem to f on $[a, b]$.

a. f must be continuous on the closed interval $[a, b]$

b. f must be differentiable on (a, b)

c. $f(a) = f(b)$

d. $f'(x) > 0$ on $[a, b]$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

8 Use the IVT to show that $x^3 + 2x - 5 = 0$ has a root in which interval?

a. $(1, 2)$

b. $(0, 1)$

c. $(2, 3)$

d. $(-1, 0)$

9 A continuous function g satisfies $g(0) = -3$ and $g(4) = 5$. Does $g(x) = \pi$ have a solution on $(0, 4)$?

a. Yes, because $-3 < \pi < 5$ and g is continuous

b. No, because π is irrational

c. Only if g is differentiable

d. Cannot be determined without $g'(x)$

Section 4: Extreme Value Theorem (EVT) and Properties

10 What does the Extreme Value Theorem guarantee for a function continuous on a closed bounded interval $[a, b]$?

a. f attains both an absolute maximum and an absolute minimum on $[a, b]$

b. f has at least one inflection point

c. f is invertible on $[a, b]$

d. $f'(c) = 0$ for some $c \in (a, b)$

11 Which function fails to achieve an absolute maximum on $(0, 1)$?

a. $f(x) = \frac{1}{x}$

b. $f(x) = x^2$

c. Both $f(x) = 1/x$ and $f(x) = x^2$

d. $f(x) = 5$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

12 If f and g are continuous at $x = c$, which of the following is NOT guaranteed to be continuous at $x = c$?

a. $f(x)/g(x)$

b. $f(x) + g(x)$

c. $f(x) \cdot g(x)$

d. $[f(x)]^2$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647